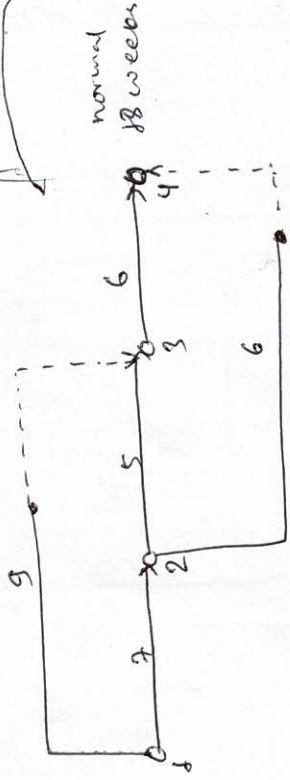


Cost slope = $\frac{\text{Graph Cost} - \text{Normal Cost}}{\text{Normal time} - \text{Graph time}}$



1 kip = 4.448 kN

$T_D = 3 \times 0.7 = 2.1$

Take $T_D = 2.5$

$1.0 < R_F = \frac{3R}{8} \leq 2$

$\frac{3R}{8} = 2.1$

Adopt $R_F = 2$

Activity	A_c	A_t	S
1-2	7500	3	2000
2-3	4500	3	15000
2-5	3000	3	1000
2-4	9000	2	3500
3-4	6000	2	3000

Elastomeric Isolators

for circular pad, $S = \frac{\phi}{4t}$ of a single layer of rubber
 ϕ = diameter
 t = thickness

$S = 10-20$ for good design

Mod. stiffness of a single isolator, $K_H = \frac{GA}{t_r}$ $\rightarrow$ shear modulus of rubber
 $\rightarrow$ full CSA of the pad
 $\rightarrow$ total thickness of rubber

Max. shear strain, $\gamma = \frac{D}{t_r}$ $\rightarrow$ max. horz. disp.

$\frac{A}{\phi} \times \gamma = \frac{\phi}{2} \times C \Rightarrow \phi = \sqrt{\frac{2A\gamma}{C}}$

$T_{OC} = N \times \gamma$
 $= \frac{A}{\phi} \times \gamma$
 $T_{CC} = \frac{\phi}{2} \times C$
 $T_{SS} = SS \times C$

Vertical stiffness of a rubber bearing,

$K_v = \frac{E_c A_s}{t_r}$ $\rightarrow$ Compression modulus of rubber-steel composite
 t_r = area of steel shim plate

For circular pad without any holes in center,

$E_c = 69 S^2$

for bearings with very large shape factors, the compressibility of rubber affects E_c , in such case more accurate $E_c = 69 S^2 k / (69 S^2 + k)$

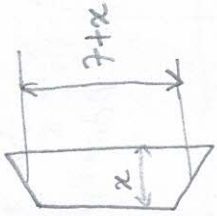
k is bulk modulus of rubber = 145 ksi to 360 ksi
 Convent = 290,000 psi

$$Req. = 14' \times 14'$$

$$Existing = 7' \times 7'$$

$$Area = 147 = 14' \times 14' - 7' \times 7'$$

$$\frac{147}{4} = 36.75$$



$$36.75 = x(7+x)$$

$$x^2 + 7x - 36.75 = 0$$

$$\Rightarrow x = 3.5$$

$$\frac{3}{2}x = 3.5 \times 1.5 = 5.25$$

$$\approx 5.5$$

$$8-20\phi$$

$$= \frac{\pi}{4} \times 20^2 \times 8$$

$$= 800\pi$$

$$Actual = 8-12\phi$$

$$= \frac{\pi}{4} \times 12^2 \times 8$$

$$= 288\pi$$

$$D = 0$$

$$\Delta/B$$

$$Area = \sqrt{\frac{\Delta \times \pi}{8}}$$

$$4/3 \text{ Area}$$

$$512\pi$$

$$64\pi$$

$$16$$

$$21.33$$

$$\approx 20$$

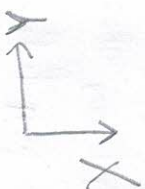
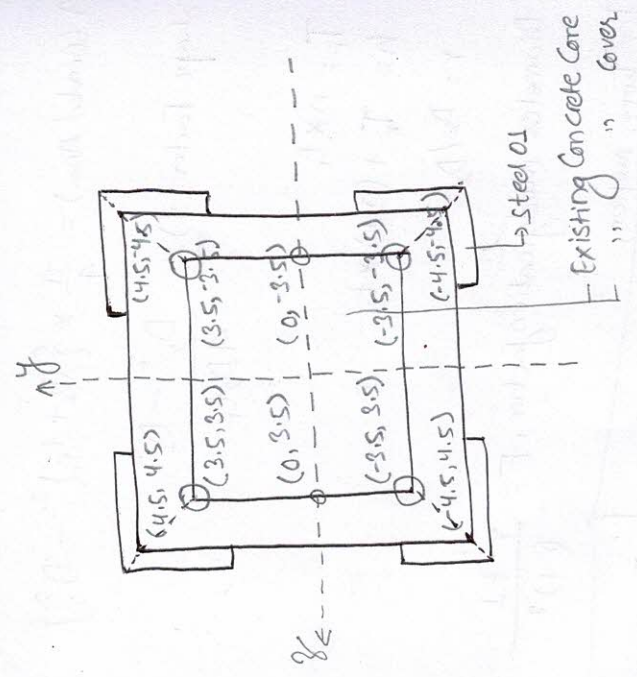
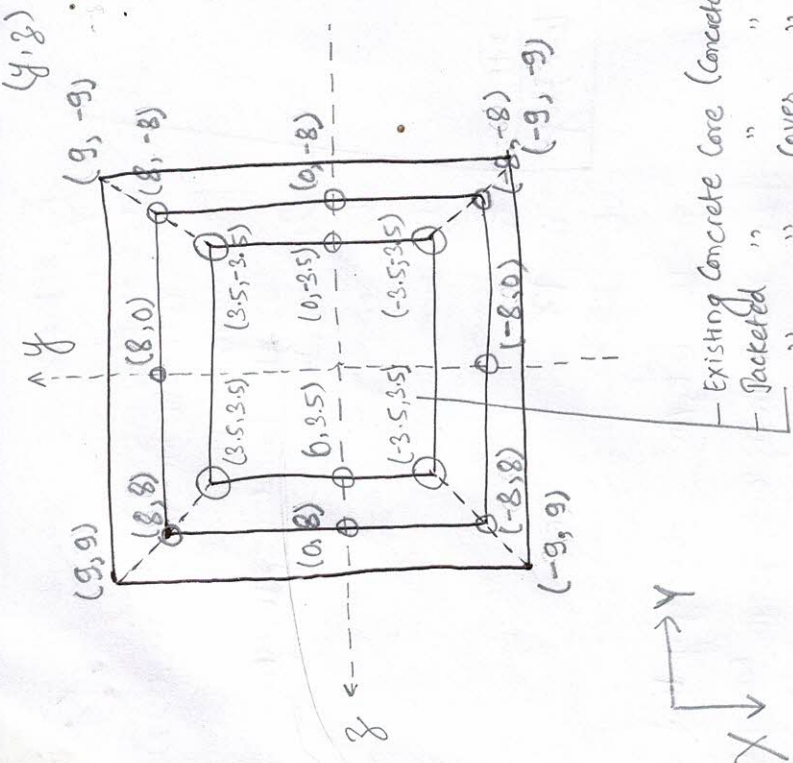
Column loads at base

11 Floor ac. (Non-Goodwood Damages)

1111 4-5 to 8 to store buildings

1111 Soil Type A

$T_{fund} \approx 22 \text{ seconds}$



Existing Concrete Core (Concrete 02)
Jacketed " " Cover "

Actual = $K A_{required}$ $\rightarrow$ = (Req. - Existing)

$K = 3/2$ for concrete

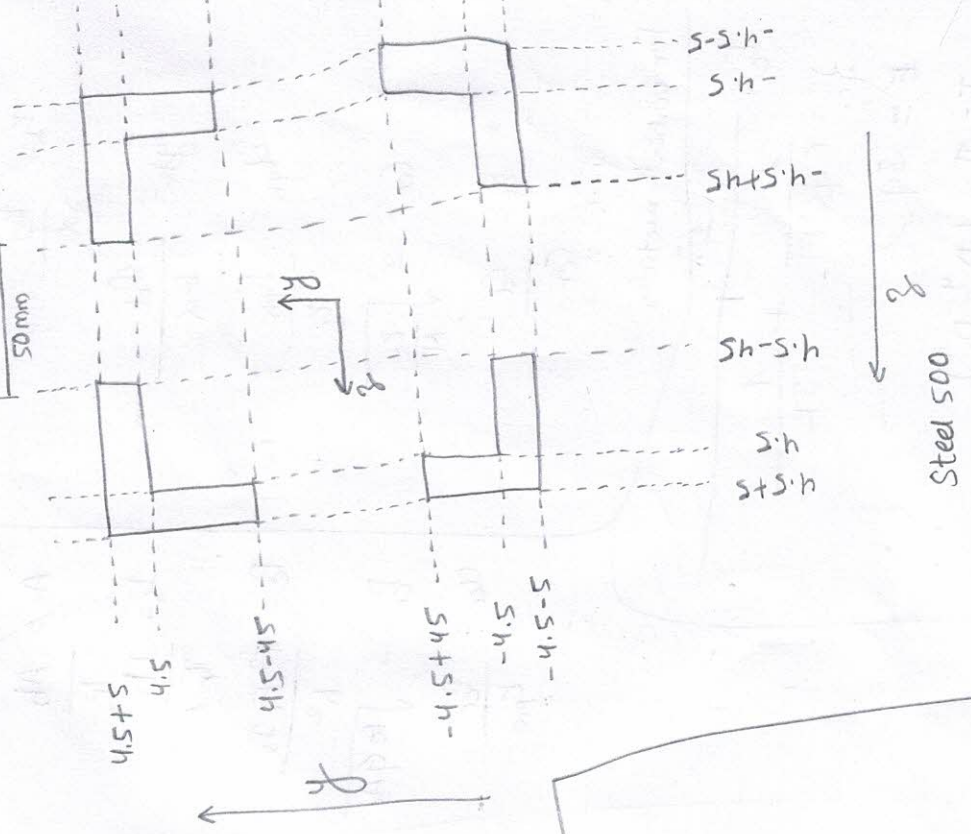
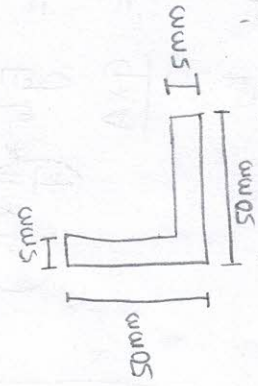
$K = 4/3$ for steel

M25 for jacketing,

Initial = 8-12 ϕ

Req $\phi = 20$

Adopt: 8-20 ϕ



Dimensions, nodes, Materials

beam-col-section

beam-col-elements

nodal-mass-and-udd

model-base

gravity-analysis

Pushover
NLTKA-individual

$$A (\text{Bonded Area}) = \frac{\pi}{4} * \{ (D_2 + t_c)^2 - D_1^2 \}$$

$$\text{Shape Factor (S)} = \frac{D_2^2 - D_1^2}{4D_2 t_r}$$

$$T_r = n * t_r$$

$$h = T_r + (n-1) * t_s$$

$$r = D_2 / D_1$$

$$\text{Diameter modification factor, } F = \frac{r^2 + 1}{(r-1)^2} + \frac{1+r}{(r-1) \log r}$$

For long motion,

$$Q_d = F_y l_0 * \left(\frac{\pi * D_1^2}{4} \right)$$

$$k_2 = \frac{G * A}{T_r}$$

$$k_1 = \frac{k_2}{\alpha}$$

$$F_{yn} = \frac{Q_d}{1-\alpha}$$

$$u_{yn} = \frac{F_{yn}}{k_1}$$

$$C_r = 2 \sqrt{\frac{k_2}{M}}$$

$$\xi_g = \frac{C_d}{C_r}$$

For vertical motion,

$$E_c = \frac{1}{\frac{1}{6GS^2F} + \frac{4}{3K}}$$

$$E = 3G$$

$$I = \frac{\pi}{64} * (D_2^4 - D_1^4)$$

$$r_g = \sqrt{\frac{I}{A}} \quad (\text{radius of gyration})$$

$$k_{pre} = \frac{AE_c}{T_r}$$

$$k_{pen} = \frac{AE}{T_r}$$

$$F_y = 19 \text{ MPa} \quad \text{yield strength}$$

$$G_r = 60 \text{ PSI} \quad \text{shear modulus}$$

$$= 413.6856 * 10^3 \text{ Pascal} \parallel 413.6856 * 10^3$$

$$K_{bulk} = 290,000 \text{ psi} \quad \text{bulk modulus of rubber}$$

$$= 2 * 10^9 \text{ Pascal} = 2 * 10 \text{ MPa}$$

$$D_1 = 3.5 \text{ inch}$$

$$D_2 = 94 \text{ inch}$$

$$t_g = 5 \text{ mm}$$

$$t_r = 10 \text{ mm}$$

$$n = 17$$

$$t_c = 15 \text{ mm}$$

$$\rho_L = 11200 \text{ kg/m}^3 \quad \text{density of lead}$$

$$c_L = 130 \text{ N-m/kg}^\circ \text{C} \quad \text{specific heat of lead}$$

$$k_S = 50 \text{ W/m}^\circ \text{C} \quad \text{thermal conductivity of steel}$$

$$a_S = 1.41 * 10^{-5} \text{ m}^2/\text{s} \quad \text{thermal diffusivity of steel}$$

$$Q_d = F_y * (1-\alpha) * \frac{\pi * D_1^2}{4}$$

$$F = ma$$

$$N = m * \text{mm/s}^2$$

$$\frac{Ns^2}{\text{mm}} = \text{mass}$$

$$1k_g = \frac{Ns^2}{m} = \frac{Ns^2}{\text{mm}} * 10^{-3}$$

$$q_L = 11200 * \left(10^{-3} \frac{Ns^2}{\text{mm}} \right) * \frac{1}{10^9 \text{ mm}^3}$$

$$= 1.12 * 10^{-8} \text{ N s}^2 / \text{mm}^4$$

$$c_L = \frac{130 * N * (10^3 \text{ mm})}{(10^{-3} \text{ N s}^2 / \text{mm})^\circ \text{C}}$$

$$= 130 * 10^6 \text{ mm}^2 / \text{s}^2 \text{ }^\circ \text{C}$$

$$k_S = 50 \left(\frac{\text{N-m}}{\text{s}} \right) \frac{1}{\text{m}^\circ \text{C}} = 50 \text{ N/s}^\circ \text{C}$$

$$a_S = 1.41 * 10^{-5} \text{ (mm}^2 * 10^6 / \text{s)} = 14.1 \text{ mm}^2 / \text{s}$$

Design drp., the drop at
center of rigidity of the isolation
system at DBE,

$$D_D = \left(\frac{g}{4\pi^2} \right) \frac{S_{DL} T_D}{B_D}$$

At MCE,

$$D_M = \left(\frac{g}{4\pi^2} \right) \frac{S_{ML} T_M}{B_M}$$

T_D, T_M = isolated periods (2.5s)

B_D, B_M = damping coefficients

S_{DL}, S_{ML} = spectral coefficients

$$S_d \approx \frac{W_D}{4D_D}$$

Ground Motions matched to soil Type A,

NLTHA at 0.35g,

LS = 2%.

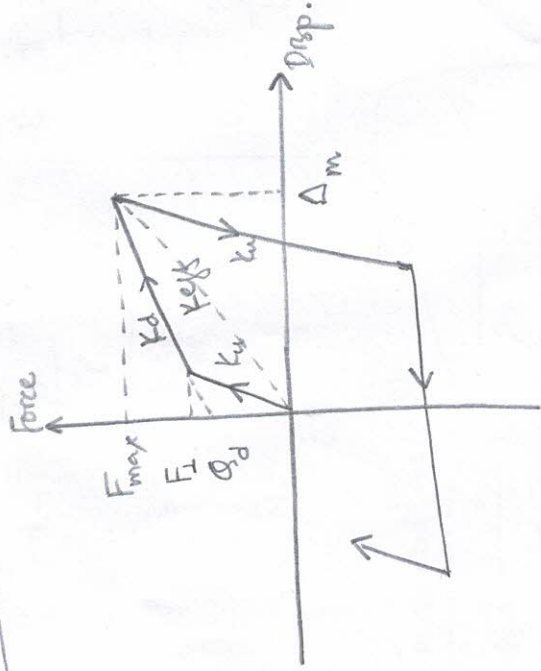
	Building Base	Obs. Drift	Peak Floor Acc.	T_L
Isolated				

Gavab Ready Mai



25-10-2020

$m \ddot{u}$



$Q_d \approx$ yield force of yield plug
yield stress of lead = 1500 psi

$$k_{eff} = k_d + \frac{Q_d}{D} \quad D \geq D_y$$

D is horz. disp. larger than yield disp.

$$T = 2\pi \sqrt{\frac{m}{k_{eff} g}}$$

$$k_u \approx 10 k_d$$

Q_d = Characteristic strength

F_y = Yield force

F_{max} = Maximum force

k_d = post-elastic stiffeners (kips/inch)

k_u = elastic (unloading) stiffeners

k_{eff} = effective stiffness of lead-play bearing

Δ_m = max. bearing displacement

eff. percentage of critical damping
provided by the isolator, P_{eff}

$$P_{eff} = \frac{4Q_d (D - Q_d / g k_u)}{2\pi (k_u D + Q_d) D}$$